Simulation Arena

Margin Protection & Floor Setting: Stress-Testing Cost Volatility

With no price escalator there is a 100.0% chance (certain - it will definitely break) this contract breaks your 12% margin floor by Year 3 - no escalator gets it back inside your 10% tolerance - this one has to be restructured.

Seed 42 | Run run_0a2ab55c42ac | 1,000 scenarios tested | Latin Hypercube Sampling of multi-year… | 22 Sep 2026, 07:30

Same seed, same numbers - this run can be reproduced exactly.

How this works

We split your cost to deliver into its moving parts and gave each one - plus the currency - a range of yearly moves, from the small ones to the nasty ones. Instead of picking randomly, we walked 1,000 complete 3-year cash flows through those ranges on an even grid, so every slice of every range gets tested and nothing important is missed. Each run asks one question: at the end of each year, is the margin still above your floor? We then repeat the whole thing at every escalator from 0% upwards until the chance of breaking the floor falls below the 10% you said you could live with. Costs are modelled with a 'bad news is more likely than good news' curve, so surprise increases weigh more than surprise windfalls.

The numbers that matter

CHANCE OF BREAKING THE 12% FLOOR BY YEAR 3

100.0%

The odds that the Year 3 margin comes in below your minimum - the final year is where cost increases have compounded the longest.

vs your tolerance is 10.0%

CHANCE OF BREAKING IT AT ANY POINT

100.0%

The odds that at least one of the years comes in below the floor.

BREAKEVEN ANNUAL ESCALATOR

none within 20.0%

No annual escalator up to 20.0% brings the risk inside your 10% tolerance. Even at the best one we tested (0.0%) the chance of breaking your floor is still 100%. A bigger escalator is not the fix here - shorten the term, reset the margin you sign at, or add a…

vs you are asking for 0.0%

EXPECTED MARGIN IN YEAR 3 (THE WORST YEAR)

-76.4%

Year 3 is where the floor is most at risk.

vs you signed at -50.0%

MARGIN AT RISK, IN TODAY'S MONEY

$1.15M

The profit you planned to book minus the profit this simulation says you will actually book, in today's money.

vs planned -$4.64M

CHANCE THE CONTRACT LOSES MONEY IN A YEAR

100.0%

How often a year comes in below zero margin, not just below the floor.

CHANCE OF BREAKING THE FLOOR IN YEAR 1

100.0%

Year 1 is the one year an escalator cannot protect - only a higher starting price can.

How we got there

1. What you signed and what it costs

You bill $3.60M and it costs you $5.41M in year 1, leaving a -50.0% margin - which is already BELOW your 12% floor, so this deal is unsignable as it stands.

Evidence: 62.0% below the floor

2. Why a fixed price is the problem

Costs do not sit still. The biggest mover is Egress & Network, which can swing 15.2% in a year, and the increases compound - year 3 carries three years of them. A frozen price carries none of that.

Evidence: 8.2% expected cost drift per year

3. Where it breaks

Across 1,000 complete cash flows, the Year 3 margin comes in below 12% in 100.0% of them, and at least one year breaks it in 100.0%. Year 3 is the weakest, with an expected margin of -76.4%. In today's money that is $1.15M of planned profit at risk.

Evidence: 100.0% chance of a breach in Year 3

4. What fixes it

No escalator up to 20% fixes this. The contract needs restructuring: a cost pass-through on Delivery Labour, a shorter term, or a higher starting price.

Evidence: escalator none within 20.0%, trigger at Year 2

Charts

Margin over the life of the contract

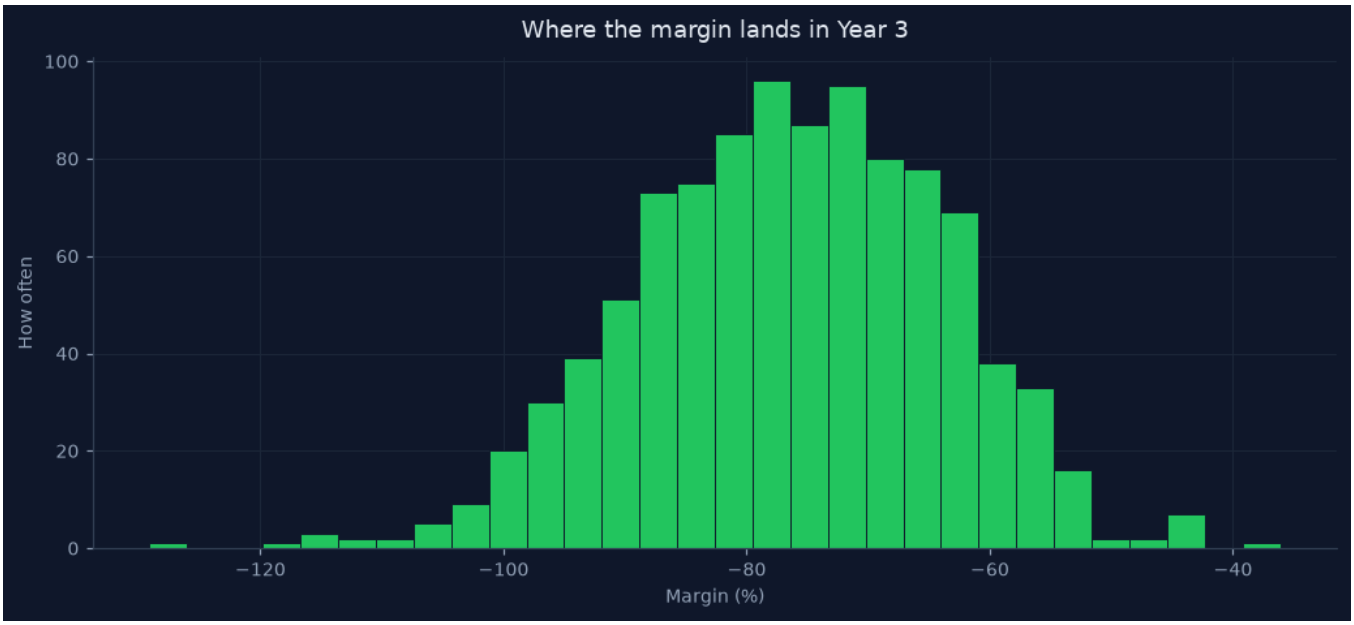
The middle line is the most likely path; the band is the range



What this shows: As long as the red line stays above the orange floor line you are safe. The moment it crosses, this contract can break your minimum margin rule.

Where the margin lands in Year 3

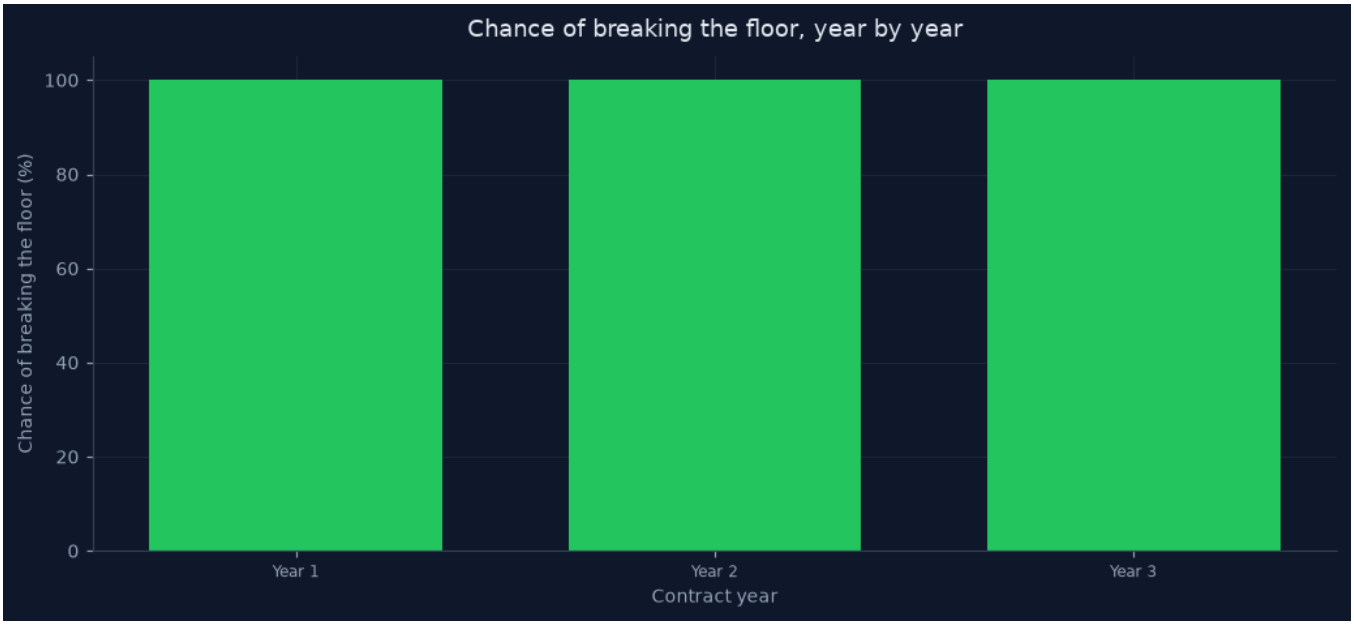
1,000 projections - the marker is your 12% floor



What this shows: Everything to the left of the floor marker is a world where this contract has broken your minimum margin rule.

Chance of breaking the floor, year by year

Your tolerance is 10% - anything above that bar is not signable as-is



What this shows: Risk builds as the contract ages because cost increases compound while a frozen price does not.

Tables

Year by year

Every one of the 1,000 projections, summarised one year at a time.

Year	You bill	Expected cost	Expected margin	A bad run (1 in 20)	A good run (1 in 20)	Chance below the floor	Chance of a loss
Year 1	$3,604,810	$5,407,215	-50.0%	-50.0%	-50.0%	100.0%	100.0%
Year 2	$3,604,810	$5,863,738	-62.7%	-78.2%	-47.7%	100.0%	100.0%
Year 3	$3,604,810	$6,359,512	-76.4%	-97.3%	-56.7%	100.0%	100.0%

Row 3 is the year most likely to break your floor.

Searching for the breakeven escalator

The same 1,000 projections re-run with a different yearly price increase. We measure the risk in Year 3, the year the cost increases have compounded the longest.

Annual escalator	Chance of breaking the floor in Year 3	Chance of breaking it at some point	Expected margin in Year 3	Margin in today's money	Verdict
0.0%	100.0%	100.0%	-76.4%	-$5.79M	too risky
0.5%	100.0%	100.0%	-74.7%	-$5.75M	too risky
1.0%	100.0%	100.0%	-72.9%	-$5.70M	too risky
1.5%	100.0%	100.0%	-71.2%	-$5.66M	too risky
2.0%	100.0%	100.0%	-69.6%	-$5.61M	too risky
2.5%	100.0%	100.0%	-67.9%	-$5.57M	too risky
3.0%	100.0%	100.0%	-66.3%	-$5.53M	too risky
3.5%	100.0%	100.0%	-64.7%	-$5.48M	too risky
4.0%	100.0%	100.0%	-63.1%	-$5.44M	too risky
4.5%	100.0%	100.0%	-61.6%	-$5.39M	too risky
5.0%	100.0%	100.0%	-60.0%	-$5.34M	too risky
5.5%	100.0%	100.0%	-58.5%	-$5.30M	too risky
6.0%	100.0%	100.0%	-57.0%	-$5.25M	too risky
6.5%	100.0%	100.0%	-55.5%	-$5.21M	too risky
7.0%	100.0%	100.0%	-54.1%	-$5.16M	too risky
7.5%	100.0%	100.0%	-52.7%	-$5.12M	too risky
8.0%	100.0%	100.0%	-51.2%	-$5.07M	too risky
8.5%	100.0%	100.0%	-49.9%	-$5.02M	too risky
9.0%	100.0%	100.0%	-48.5%	-$4.98M	too risky
9.5%	100.0%	100.0%	-47.1%	-$4.93M	too risky
10.0%	100.0%	100.0%	-45.8%	-$4.88M	too risky
10.5%	100.0%	100.0%	-44.5%	-$4.84M	too risky
11.0%	100.0%	100.0%	-43.2%	-$4.79M	too risky
11.5%	100.0%	100.0%	-41.9%	-$4.74M	too risky
12.0%	100.0%	100.0%	-40.6%	-$4.69M	too risky

Showing the first 25 of 41 rows. No row reaches 'inside tolerance'. The highlighted row is the best we found, and it is still too risky - so the escalator is not the lever that fixes this contract.

What goes into the model

Cost mix from 4 cost lines observed across 5 years in your file.

Cost line	Share of cost	Year 1 cost	Expected drift per year	Volatility	Share of the Year 3 wobble	Expected extra cost by Year 3
Delivery Labour	44.4%	$2,399,396	11.3%	8.4%	27.0%	$586,015
Cloud Hosting	41.4%	$2,236,397	4.9%	5.2%	7.1%	$225,971
Third-party Licensing	11.1%	$599,518	7.0%	12.0%	3.0%	$91,736
Egress & Network	3.2%	$171,905	11.0%	15.2%	0.5%	$42,370
Currency	-	-	0%	6.0%	62.5%	$11,446

The bottom line

What you planned versus what the simulation says, in today's money.

Column 1	Planned (margin held flat)	What the simulation says	Difference
Margin in Year 1	-50.0%	-50.0%	0.0%
Margin in Year 3	-50.0%	-76.4%	-26.4%
Margin in today's money	-$4.64M	-$5.79M	-$1.15M

Column 1	Planned (margin held flat)	What the simulation says	Difference
Chance of breaking the floor in Year 3	0%	100.0%	100.0%
Chance of breaking it at some point	0%	100.0%	100.0%

Row 3 is the value at risk from cost volatility.

What we recommend

Restructure before signing: a 100.0% chance of breaking the floor cannot be fixed by an escalator alone.

Risk High 100 out of 100 | Confidence: High

Why

• Expected margin falls from -50.0% in Year 1 to -76.4% by Year 3.

• Chance of breaking the 12% floor by Year 3: 100.0% against a tolerance of 10.0% (100.0% if you count every year).

• Margin at risk, in today's money: $1.15M of the -$4.64M you planned to book.

What to do

1. Raise the starting margin or shorten the term - the gap between -50.0% and your 12% floor is too thin for this level of cost volatility.

2. Move Delivery Labour to a pass-through or a capped index; it drives 27.0% of the uncertainty.

3. Add a hard renegotiation trigger at the first year the margin would break the floor (Year 2).

The trade-off: Every point of escalator you win protects about $0 of what that money is worth today - and every point you ask for is a point buyer will not wear the escalator, take the renegotiation trigger instead: it costs nothing upfront and gives you an exit.

What we assumed and what to watch

Worth knowing before you act

• With no escalator there is roughly a certain chance of breaking your 12% floor by Year 3 - that is the whole risk of a frozen price.

• There is a 100.0% chance of an outright loss-making year, not just a breach of the floor.

• Your file shows Third-party Licensing, Egress & Network moving too smoothly to measure a real swing from, so we kept your volatility setting for them rather than reading near-zero risk off a short history.

• Your file reports an average margin of -50.0%, so that - not the 15.0% you typed - sets year-1 cost. If you meant the figure you typed, clear the margin column from the mapping and run again.

• Contract value was read from your file ($3.60M average), overriding the $8.00M you typed.

• Your file holds 720,000 rows. We are running on an evenly spread sample of 239,963 of them (33.3%), keeping whole records (grouped by your 'fx_spot_rate_realized' column) intact rather than splitting a record down the middle, because a working set that big does not fit in memory while keeping the tool responsive. Every number below comes from those 239,963 rows.

Assumptions

• Contract: $3,604,810 billed per year for 3 years, starting margin -50.0%, so year-1 cost is $5,407,215.

• Cost mix: 4 cost lines observed across 5 years in your file - Delivery Labour 44.4% of cost, drift 11.3% per year, volatility 8.4%, Cloud Hosting 41.4% of cost, drift 4.9% per year, volatility 5.2%, Third-party Licensing 11.1% of cost, drift 7.0% per year, volatility 12.0%, Egress & Network 3.2% of cost, drift 11.0% per year, volatility 15.2%.

• Currency: a separate wobble of 6.0% applied to the whole cost base.

• Capacity cliff: none modelled (set the step jump above 0% to include one).

• Attrition: none modelled (set a backfill cost above 0% to include it).

• Each cost line follows a 'bad news weighs more than good news' curve: the middle outcome is the drift, but surprise increases are larger than surprise windfalls.

• Year 1 is priced on the cost you can see today; every year after that picks up a fresh cost shock over 2 year(s).

• Each volatility figure is read as the total swing for that cost line across the shocked years and spread evenly over them, so a 15.2% figure means 'ends up within about that much of plan by Year 3', not 'that much every single year'. Costs still compound - good and bad years stack up. Year-on-year volatility learned from your file is floored at 4%, because we will not assume a cost line is steadi…

• Cost lines in the same year move together somewhat (they share the same economy); years move independently of each other.

• Margin floor 12%, risk tolerance 10%, discount rate 8.0% used to value future money, escalator in this run 0.0%.

• 1,000 complete cash flows on an even grid; seed 42 - re-running gives identical numbers.

• Contract ID -> your 'contract_id' column

• Customer name -> your 'customer' column

• Contract value (revenue) -> your 'contract_value' column

• Year -> your 'year' column

• Cost component -> your 'cost_component' column

• Cost in that year -> your 'base_cost' column

• Margin achieved (%) -> your 'margin_pct' column

• Capacity cliff flag -> your 'capacity_cliff_flag' column

• Staff leavers (count) -> your 'attrition_turnover_count' column

Appendix: the last steps of the run

Every line was produced while the model ran. Re-running with the same seed replays it exactly.

Time	Stage	What happened
1.38s	prepare	Cost component -> your 'cost_component' column
1.38s	prepare	Cost in that year -> your 'base_cost' column
1.38s	prepare	Margin achieved (%) -> your 'margin_pct' column
1.38s	prepare	Capacity cliff flag -> your 'capacity_cliff_flag' column
1.38s	prepare	Staff leavers (count) -> your 'attrition_turnover_count' column
7.21s	prepare	Read 239,963 records into the model.
7.21s	simulate	Running the simulation…
7.21s	read	Reading the contract and how its cost is built up...
7.68s	read	Learned the cost mix and its drift from your file: 4 cost lines across 5 years.
7.68s	read	Your file shows 2 cost line(s) moving too smoothly to measure a swing from (Third-party Licensing, Egress & Network), so we kept the volatility you typed in for them.
7.74s	read	Your file reports an average starting margin of -50.0%, so we used that rather than the default.
7.80s	read	Year 1: you bill $3.60M and it costs you $5.41M - a -50.0% margin.

Time	Stage	What happened
7.80s	simulate	Laying out 1,000 complete 3-year cash flows across every possible path your costs could take...
7.82s	simulate	Walked 10% of the cash flows...
7.82s	simulate	Walked 20% of the cash flows...
7.82s	simulate	Walked 40% of the cash flows...
7.82s	simulate	Walked 50% of the cash flows...
7.82s	simulate	Walked 60% of the cash flows...
7.82s	simulate	Walked 70% of the cash flows...
7.82s	simulate	Walked 80% of the cash flows...
7.82s	simulate	Walked 90% of the cash flows...
7.94s	summarize	Reading the results year by year...
7.94s	summarize	Solving for the annual escalator that brings the risk back inside your tolerance...
7.94s	summarize	No annual escalator up to 20.0% brings the risk inside your 10% tolerance. Even at the best one we tested (0.0%) the chance of breaking your floor is still 100%. A bigger escalator is not the fix here - shorten the term…
7.95s	summarize	Done. 100.0% chance of breaking the floor by Year 3; no escalator fixes it - this one needs restructuring.